Topology and Manifolds
MH3600
Chapter 6: Compactness

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Outline

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Definition

Definition

A collection $\mathcal{A}$ of subsets of a space X is a covering of X (or is said to cover X), if the union of the elements of $\mathcal{A}$ is equal to X. The covering $\mathcal{A}$ is an open cover of X if its elements are open subsets of X. If Y is a subspace of X, a collection $\mathcal{A}$ of subsets of X covers Y if the union of its elements contains Y.

Definition

A space X is compact if every open cover $\mathcal{A}$ of X contains a finite subcollection that also covers X (called a finite subcover).

Examples

Example

The real line $\mathbb{R}$ is not compact since the open covering $\mathcal{A} = \{(n, n+2) \mid n \in \mathbb{Z}\}$ has no finite subcover since any finite subset of $\mathcal{A}$ can contain only a finite number of elements of $\mathbb{Z}$.

Example

The bounded open interval $(a, b) \subset \mathbb{R}$ is not compact since the open covering $\mathcal{A} = \{(a + 1/n, b - 1/n) \mid n \in \mathbb{N}, 1/n < (b-a)/2\}$ has no finite subcover, since any finite subset of $\mathcal{A}$ only covers $(a + 1/N, b - 1/N)$ for some $N \in \mathbb{N}$.

Examples

Example

Consider $X = \{0\} \cup \{1/n \mid n \in \mathbb{N}\} \subset \mathbb{R}$. Given any open covering $\mathcal{A}$ of X, there is an element U of $\mathcal{A}$ containing 0. The open set U contains all but finitely many of the points $1/n$. For each point of X not in U, choose an element of $\mathcal{A}$ containing the point. Then U, along with these finite number of elements of $\mathcal{A}$, form a finite subcover of X. So X is compact.

Example

Any space X containing only finitely many points is necessarily compact.

Note

Notice that to show a space is not compact, you can just find an open covering with no finite subcover. However, to show a space *is* compact by definition, you must consider all possible open coverings.

Lemma

Let Y be a subspace of X. Then Y is compact if and only if every covering of Y by sets open in X contains a finite subcollection covering Y.

Proof

Suppose that Y is compact and $\mathcal{A} = \{A_\alpha\}_{\alpha \in J}$ is a covering of Y by sets open in X. Then the collection $\{A_\alpha \cap Y \mid \alpha \in J\}$ is a covering of Y by sets open in Y. Since Y is compact, there is a finite subcollection $\{A_{\alpha_1} \cap Y, A_{\alpha_2} \cap Y, \ldots, A_{\alpha_n} \cap Y\}$ covering Y. Then $\{A_{\alpha_1}, A_{\alpha_2}, \ldots, A_{\alpha_n}\}$ is a finite subcollection of $\mathcal{A}$ that covers Y.

Proof (continued)

Conversely, suppose every covering of Y by sets open in X contain a finite subcollection covering Y. Let $\mathcal{A}' = \{A'_\alpha\}$ be an arbitrary covering of Y by sets open in Y. For each α, choose a set A_α open in X such that $A'_\alpha = A_\alpha \cap Y$ (this can be done since Y has the subspace topology and A'_α is open in Y). The collection $\mathcal{A} = \{A_\alpha\}$ is a covering of Y by sets open in X. Then by the hypothesis, some finite subcollection $\{A_{\alpha_1}, A_{\alpha_2}, \ldots, A_{\alpha_n}\}$ covers Y. Then $\left\{A'_{\alpha_1}, A'_{\alpha_2}, \ldots, A'_{\alpha_n}\right\}$ is a subcollection of $\mathcal{A}'$ that covers Y. Therefore, Y is compact. ■

Theorem

Every closed subspace of a compact space is compact.

Proof

Let Y be a closed subspace of the compact set X. Let $\mathcal{A}$ be an arbitrary open cover of Y by sets open in X. Let $\mathcal{B} = \mathcal{A} \cup \{X \backslash Y\}$. Then $\mathcal{B}$ is a covering of X by open sets and since X is compact then some finite subcollection of $\mathcal{B}$ covers X. This finite subcollection with $X \backslash Y$ removed (if $X \backslash Y$ is in the subcollection) is then a finite subcollection of $\mathcal{A}$ which covers Y. So by the preceding Lemma, Y is compact. ■

Compact Hausdorff spaces

The next lemma and theorem are properties of compact Hausdorff spaces.

Lemma

If Y is a compact subspace of the Hausdorff space X and $x_0 \notin Y$, then there exist disjoint open sets U and V of X, such that $x_0 \in U$ and $Y \subset V$.

Proof

Since X is Hausdorff, then for each $y \in Y$ there are disjoint open U_y and V_y with $x_0 \in U_y$ and $y \in V_y$. Then $\{V_y \mid y \in Y\}$ is a covering of Y by sets open in X. Since Y is hypothesized to be compact, then by the preceding Lemma there are finitely many elements of the covering which covers Y, say $V_{y_1}, V_{y_2}, \ldots, V_{y_n}$. Define $V = V_{y_1} \cup V_{y_2} \cup \cdots \cup V_{y_n}$ and $U = U_{y_1} \cap U_{y_2} \cap \cdots \cap U_{y_n}$. Then U and V are open, U and V are disjoint, $x_0 \in U$ and $Y \subset V$, as claimed. ∎

Theorem

Every compact subspace of a Hausdorff space is closed.

Proof

Let Y be a compact subspace of Hausdorff space X. Let $x_0 \in X \backslash Y$. Then by the preceding Lemma there is an open subset $U \subset X \backslash Y$ with $x_0 \in U$. Therefore x_0 is an interior point of $X \backslash Y$ (by definition of interior of a set) and so $X \backslash Y = \mathrm{int}(X \backslash Y)$ and hence $X \backslash Y$ is open and therefore Y is closed. ■

We can use this Theorem to show that some subspaces are not compact by showing that they are not closed.

Example

Since $\mathbb{R}$ is Hausdorff, we know that any nonclosed subset of $\mathbb{R}$ is not compact. For example, neither $(a, b]$, $[a, b)$, nor (a, b) are compact.

Example

The Hausdorff condition is necessary in the previous Theorem.
Consider the finite complement topology on $\mathbb{R}$ given by

$$\mathcal{T} = \{U \subset \mathbb{R} \mid \mathbb{R}\backslash U \text{ is finite or all of } \mathbb{R}\}$$

(similar to Prob. 2 of Tutorial 2). So the only closed sets are the finite sets and $\mathbb{R}$. But it can be shown that every subset of $\mathbb{R}$ is compact in this topology so there are compact subspaces which are not closed. This is possible because the finite complement topology on $\mathbb{R}$ is not Hausdorff.

Compactness and continuity

Theorem

The image of a compact space under a continuous map is compact.

Proof

Let $f : X \to Y$ be continuous and X compact. Let $\mathcal{A}$ be an arbitrary covering of $f(X)$ by sets open in Y. Since f is continuous with domain X, the collection $\left\{f^{-1}(A) \mid A \in \mathcal{A}\right\}$ is a collection of open sets in X which covers X. Since X is compact, then there is finite subcollection $f^{-1}(A_1), f^{-1}(A_2), \ldots, f^{-1}(A_n)$ which covers X. But then the finite subcollection $\{A_1, A_2, \ldots, A_n\}$ of $\mathcal{A}$ covers $f(X)$. So $f(X)$ is compact.
■

Theorem

Let $f : X \to Y$ be a bijective continuous function. If X is compact and Y is Hausdorff, then f is a homeomorphism.

Proof

Since f is a bijection, then $f^{-1} : Y \to X$ is defined. Let $A \subset X$ be closed. Then by Theorem on slide 9, A is compact. By Theorem on slide 14, $f(A)$ is compact. Since Y is Hausdorff, by Theorem on slide 11, $f(A)$ is closed. So f^{-1} is continuous by a Theorem of Chapter 2. So f is a continuous bijection with a continuous inverse; that is, f is a homeomorphism. ■

Compactness and product spaces

The following lemma allows us to address the compactness of a product of compact spaces.

Lemma (The Tube Lemma)

Consider the product space $X \times Y$ where Y is compact. If N is an open set of $X \times Y$ containing the slice $\{x_0\} \times Y$ of $X \times Y$, then N contains some "tube" $W \times Y$ about $\{x_0\} \times Y$, where W is a neighborhood of x_0 in X.

Proof of the Tube Lemma

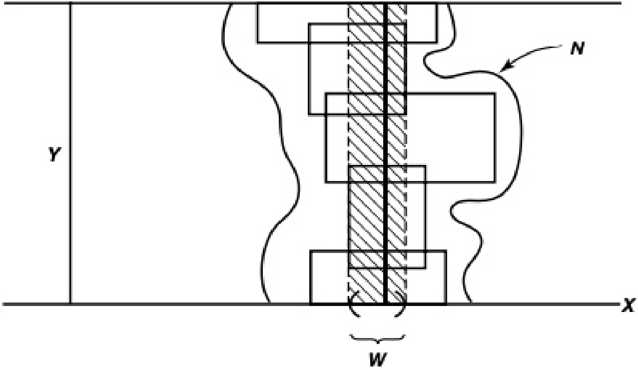
First, each element $z \in \{x_0\} \times Y$ is an element of some basis element of the product topology. Since N is open and $z \in N$, then z is in some basis element, which is a subset of N by Lemma slide 15 of Chap. 2. So z is in a basis element $U \times V$ of the product topology which is a subset of N (see part (2) of the definition of basis).

Let X be a set and let $\mathcal{B}$ be a basis for a topology $\mathcal{T}$ on X. Then $\mathcal{T}$ equals the collection of all unions of elements of $\mathcal{B}$.

Proof of the Tube Lemma (continued)

Second, since $\{x_0\} \times Y$ is homeomorphic to Y and Y is compact, then $\{x_0\} \times Y$ is compact. So $\{x_0\} \times Y$ can be covered by finitely many of these sets, say $U_1 \times V_1, U_2 \times V_2, \ldots, U_n \times V_n$. Without loss of generality, each of these sets intersects $\{x_0\} \times Y$ (otherwise, they can be eliminated from the covering). Define $W = U_1 \cap U_2 \cap \cdots \cap U_n$, so that W is open and $x_0 \in W$.

Finally, let $(x, y) \in W \times Y$. Then $y \in V_{i'}$ for some $i' \in \{1, 2, \ldots, n\}$. But $x \in W$ and so $x \in U_i$ for all $i \in \{1, 2, \ldots, n\}$. Therefore $(x, y) \in U_{i'} \times V_{i'}$. So $W \times Y \subset \cup_{i=1}^{n} U_i \times V_i \subset N$. So $W \times Y$ is the "tube" claimed to exist. $\blacksquare$



Taken from [Munkres]

Theorem

The product of finitely many compact spaces is compact.

Proof

We prove the result for two spaces and then the general result follows by induction. Let X and Y be compact spaces and let $\mathcal{A}$ be an open covering of $X \times Y$. Given $x_0 \in X$, the slice $\{x_0\} \times Y$ is compact since it is homeomorphic to Y. Hence $\{x_0\} \times Y$ can be covered by a finite number of elements of $\mathcal{A}$, say $A_1, A_2, \ldots, A_m$. Then $N = A_1 \cup A_2 \cup \cdots \cup A_m$ is an open set containing $\{x_0\} \times Y$. By The Tube Lemma, there is a tube $W \times Y \subset N$ containing $\{x_0\} \times Y$ where W is open in X. So $W \times Y$ is covered by $A_1, A_2, \ldots, A_m$ of $\mathcal{A}$.

Proof (continued)

Thus for each $x \in X$, there is W_x a neighborhood of x such that the tube $W_x \times Y$ can be covered by finitely many elements of $\mathcal{A}$. Now the collection of all such W_x, $\{W_x \mid x \in X\}$, is an open covering of X; since X is compact, there is a finite subcollection $\{W_1, W_2, \ldots, W_k\}$ covering X. Then the union of tubes $W_1 \times Y, W_2 \times Y, \ldots, W_k \times Y$ is all of $X \times Y, X \times Y = \cup_{i=1}^k W_i \times Y$. Since each $W_i \times Y$ can be covered by finitely many elements of $\mathcal{A}$, then $X \times Y$ can be covered by finitely many elements of $\mathcal{A}$. Hence $X \times Y$ is compact and the result follows. ∎

In fact, a product of an arbitrary number of compact spaces is compact. This result is called the Tychonoff Theorem (later).

Compactness using closed sets

Definition

A collection $\mathcal{C}$ of subsets of X has the finite intersection property if for every finite subcollection $\{C_1, C_2, \ldots, C_n\}$ of $\mathcal{C}$, the intersection $C_1 \cap C_2 \cap \cdots \cap C_n$ is nonempty.

Theorem

Let X be a topological space. Then X is compact if and only if for every collection $\mathcal{C}$ of closed sets in X having the finite intersection property, the intersection $\cap_{C \in \mathcal{C}} C$ for all elements of $\mathcal{C}$ is nonempty.

Proof

Given a collection $\mathcal{A}$ of subsets of X, let $\mathcal{C} = \{X \backslash A \mid A \in \mathcal{A}\}$. Then the following hold:

(1) $\mathcal{A}$ is a collection of open sets if and only if $\mathcal{C}$ is a collection of closed sets.

(2) The collection $\mathcal{A}$ covers X ($X = \cup_{A \in \mathcal{A}} A$) if and only if the intersection $\cap_{C \in \mathcal{C}} C$ of all elements of $\mathcal{C}$ is empty (since each $x \in X$ must be in some $A \in \mathcal{A}$ and so $x \notin X \backslash A = C \in \mathcal{C}$).

(3) The finite subcollection $\{A_1, A_2, \ldots, A_n\} \subset \mathcal{A}$ covers X if and only if the intersection of the corresponding elements $C_i = X \backslash A_i$ of $\mathcal{C}$ is empty.

Proof (continued)

The statement that X is compact is equivalent to:

> *"Given any collection $\mathcal{A}$ of open subsets of X, if $\mathcal{A}$ covers X then some finite subcollection of $\mathcal{A}$ covers X."*

The (logically equivalent) contrapositive of this statement is:

> *"Given any collection $\mathcal{A}$ of open subsets of X, if no finite subcollection of $\mathcal{A}$ covers X, then $\mathcal{A}$ does not cover X."*

This second statement can be restated using (1), (2), and (3) as:

> *"Given any subcollection $\mathcal{C}$ of closed sets [by (1)], if every finite intersection of elements of $\mathcal{C}$ is nonempty [by (3)], then the intersection of all the elements of $\mathcal{C}$ is nonempty [by (2)]."*

So the property of compactness of X is equivalent to the property involving the collection of closed sets.

(1) $\mathcal{A}$ is a collection of open sets if and only if $\mathcal{C}$ is a collection of closed sets.
(2) The collection $\mathcal{A}$ covers X ($X = \cup_{A\in\mathcal{A}} A$) if and only if the intersection $\cap_{C\in\mathcal{C}} C$ of all elements of $\mathcal{C}$ is empty.
(3) The finite subcollection $\{A_1, A_2, \ldots, A_n\} \subset \mathcal{A}$ covers X if and only if the intersection of the corresponding elements $C_i = X\backslash A_i$ of $\mathcal{C}$ is empty.

Corollary

Let X be a compact topological space and let $C_1 \supset C_2 \supset \cdots \supset C_n \supset C_{n+1} \supset \cdots$ be a nested sequence of closed sets in X. If each C_n is nonempty, then $\cap_{n\in\mathbb{N}} C_n$ is nonempty.

Proof

For any finite collection of sets in $\mathcal{C}$, we have that the intersection equals C_N for some $N \in \mathbb{N}$, since the sets are nested, and $C_N \neq \varnothing$. So $\mathcal{C}$ has the finite intersection property. Since X is compact then, by the previous theorem, $\cap_{n\in\mathbb{N}} C_n$ is nonempty. $\blacksquare$

Recall: A collection $\mathcal{C}$ of subsets of X has the finite intersection property if for every finite subcollection $\{C_1, C_2, \ldots, C_n\}$ of $\mathcal{C}$, the intersection $C_1 \cap C_2 \cap \cdots \cap C_n$ is nonempty.

Recall: X is compact if and only if for every collection $\mathcal{C}$ of closed sets in X having the finite intersection property, the intersection $\cap_{C\in\mathcal{C}} C$ for all elements of $\mathcal{C}$ is nonempty.

Compactness using subbases

In Chapter 1 we introduced the notions of basis and subbasis using open sets. An analogous concept can be introduced using closed sets.

Definition

Let X be a topological space. A class of closed subsets of X is called a closed basis if the class of all complements of its sets is an open basis, and a closed subbasis if the class of all complements is an open subbasis.

Important property

Since the class of all finite intersections of sets in an open subbasis is an open basis, it follows that the class of all finite unions of sets in a closed subbasis is a closed basis. This is called the closed basis generated by the closed subbasis.

The proof of the following theorem is quite convoluted and uses Zorn's Lemma: In a partially ordered set (comparability, nonreflexivity, transitivity), if every chain has an upper bound, then the set contains at least one maximal element.
It can be found in [Simmons].

Theorem

A topological space is compact if every subbasic open cover has a finite subcover, or equivalently (see slide 23), if every class of subbasic closed sets with the finite intersection property has non-empty intersection.

Tychonoff's Theorem

Theorem

The product of any non-empty class of compact spaces is compact.

Note: this uses the product (not box) topology!

Proof of Tychonoff's Theorem

Let $\{X_\alpha\}_{\alpha\in I}$ be a non-empty class of compact spaces, and form the Cartesian product $X = \prod_{\alpha\in I} X_\alpha$.
Let $\left\{F_j\right\}_{j\in J}$ be a non-empty subclass of the defining closed subbasis for the product topology on X. This means that each F_j is a product of the form $F_j = \prod_{\alpha\in I} F_{\alpha j}$, where $F_{\alpha j}$ is a closed subset of X_α which equals X_α for all $\alpha \in I$ but one.

Recall: a subbasis for the product topology consists of subsets of the form $\prod_{\alpha\in I} U_\alpha$, where $U_\alpha \subset X_\alpha$ is an open subset which equals X_α for all $\alpha \in I$ but one.

We assume that the class $\left\{F_j\right\}_{j\in J}$ has the finite intersection property, and by virtue of the Theorem in slide 26, to conclude the proof we must show that $\bigcap_{j\in J} F_j$ is non-empty. For a given fixed $\alpha, \left\{F_{\alpha j}\right\}_{j\in J}$ is a class of closed subsets of X_α with the finite intersection property; and by the assumed compactness of X_α (and Theorem in slide 21), there exists a point x_α in X_α which belongs to $\cap_j F_{\alpha j}$. If we do this for each α, we obtain a point $x = (x_\alpha)_{\alpha\in I}$ in X which is in $\cap_j F_j$. $\blacksquare$

Outline

Compact Subspaces of the Real Line

We shall prove that every closed interval in $\mathbb{R}$ is compact. Applications include the extreme value theorem and the uniform continuity theorem of calculus, suitably generalized. We also give a characterization of all compact subspaces of $\mathbb{R}^n$, and a proof of the uncountability of the set of real numbers.

To prove every closed interval in $\mathbb{R}$ is compact, we need only one of the order properties of the real line: the least upper bound property. We shall prove the theorem using only this hypothesis; then it will apply not only to the real line, but to well-ordered sets and other ordered sets as well.

Compact subsets of simply ordered sets

Theorem

Let X be a simply ordered set having the least upper bound property. In the order topology, each closed interval X is compact.

Recall: An ordered set A is said to have the least upper bound property if every nonempty subset A_0 of A that is bounded above has a least upper bound (i.e., the set of all upper bounds for A_0 has a smallest element).

Proof

Let $a < b$ and let $\mathcal{A}$ be a covering of a, b by sets open in $[a, b]$ in the subspace topology (which is the same as the order topology).

- **Step 1**. We show that for each $x \in [a, b)$, there is $y \in [a, b]$, $y > x$ such that $[x, y]$ can be covered by at most two elements of $\mathcal{A}$.
  Let $x \in [a, b)$.
  – If x has an immediate successor in X, let y be this immediate successor. Then $[x, y] = \{x, y\}$ and $[x, y]$ can be covered by at most two elements of $\mathcal{A}$.
  – If x has no immediate successor in X, choose an element $A \in \mathcal{A}$ containing x. Because $x \neq b$ and A is open, A contains an interval of the form $[x, z)$ for some $z \in [a, b]$. Choose $y \in (x, z)$. Then the interval $[a, y]$ is covered by the single element A of $\mathcal{A}$.

Recall: y immediate successor of x if (x, y) is empty (no z such that $x < z < y$).

Proof (continued)

- **Step 2**. Let $C = \{y \in (a, b] \mid [a, y]$ can be covered by finitely many elements of $\mathcal{A}\}$. Since $a \in C, C \neq \varnothing$.
  Let c be the least upper bound of set C (this is where the least upper bound property is used).
  Then, by Step 1 (with $x = a$) 1, $a < c \leq b$.
- **Step 3**. We show that $c \in C$. Since $\mathcal{A}$ is a covering of $[a, b]$, then some $A \in \mathcal{A}$ contains c. Since A is open, it contains an interval of the form $(d, c]$ for some $d \in [a, b]$.
  If $c \notin C$ (the least upper bound of C is not in C), there must be a point z of C lying in the interval (d, c), because otherwise d would be a smaller upper bound on C than c.
  Since $z \in C$, the interval $[a, z]$ can be covered by finitely many (say n) elements of $\mathcal{A}$ (by the definition of C).
  Now $[z, c] \subset (d, c] \subset A \in \mathcal{A}$, hence $[a, c] = [a, z] \cup [z, c]$ can be covered by $n + 1$ elements of $\mathcal{A}$. But then $c \in C$, a contradiction. So the assumption that $c \notin C$ is false, and in fact $c \in C$.

Proof (last step)

- **Step 4**. We show that $c = b$. Assume $c < b$.
  By Step 1 with $x = c$ (possible since $c < b$), there is $y \in [a, b]$ with $y > c$ such that $[c, y]$ can be covered by finitely many elements of $\mathcal{A}$.
  From Step 3, $c \in C$ and so $[a, c]$ can be covered by finitely many elements of $\mathcal{A}$.
  So, $[a, y] = [a, c] \cup [c, y]$ can be covered by finitely many elements of $\mathcal{A}$, so $y \in C$. But $y > c$, contradicting the fact that c is an upper bound of C. So the assumption that $c < b$ is false and so $c = b$ (notice $c \leq b$ by Step 2).
  By Step 3, $b = c \in C$ and so the interval $[a, b]$ can be covered by finitely many elements of $\mathcal{A}$. Since $\mathcal{A}$ is an arbitrary open covering of $[a, b]$, then $[a, b]$ is compact. ■

Corollary

Every closed and bounded interval $[a, b] \subset \mathbb{R}$ *(where* $\mathbb{R}$ *has the standard topology) is compact.*

The Heine-Borel Theorem

The following theorem classifies compact subspaces of $\mathbb{R}^n$ under the Euclidean metric (and under the square metric). Recall that the standard topology on $\mathbb{R}$ is the same as the metric topology under the Euclidean metric, so this theorem includes and extends the previous corollary. We call this the Heine-Borel Theorem.

Theorem (The Heine-Borel Theorem)

A subspace A of $\mathbb{R}^n$ is compact if and only if it is closed and is bounded in the Euclidean metric d or the square metric ρ.

Recall:

– Euclidean metric on $\mathbb{R}^n$:

$$d(x, y) = \left((x_1 - y_1)^2 + (x_2 - y_2)^2 + \cdots + (x_n - y_n)^2\right)^{1/2}$$

– square metric on $\mathbb{R}^n$:

$$\rho(x, y) = \max\{|x_1 - y_1|, |x_2 - y_2|, \ldots, |x_n - y_n|\}$$

Proof

Recall from Theorem slide 39 of Chap. 3 that the topologies on $\mathbb{R}^n$ induced by d and ρ are the same as the product topology (and the box topology) on $\mathbb{R}^n$.
We have the inequality $\rho(\mathbf{x},\mathbf{y}) \leq d(\mathbf{x},\mathbf{y}) \leq \sqrt{n}\rho(\mathbf{x},\mathbf{y})$ for all $\mathbf{x},\mathbf{y} \in \mathbb{R}^n$, so that A is bounded under d if and only if A is bounded under ρ.

(1) Suppose that A is compact. Since $\mathbb{R}^n$ is Hausdorff, by the Theorem on slide 11, A is closed. Consider the collection of open sets $\{B_\rho(\mathbf{0},m) \mid m \in \mathbb{N}\}$, whose union is all of $\mathbb{R}^n$. Since A is compact, some finite subcollection covers A and so $A \subset B_\rho(\mathbf{0},M)$ for some $M \in \mathbb{N}$. So (by the Triangle Inequality) for any $\mathbf{x},\mathbf{y} \in A$ we have $\rho(\mathbf{x},\mathbf{y}) \leq 2M$ and hence A is bounded under ρ. That is, A is closed and bounded.

Proof

(2) Conversely, suppose that A is closed and bounded under ρ, say $\rho(\mathbf{x}, \mathbf{y}) \leq N$ for all $\mathbf{x}, \mathbf{y} \in A$. Let $\mathbf{x}_0 \in A$ be given and define $b := \rho(\mathbf{x}_0, \mathbf{0})$. For every $\mathbf{x} \in A$, we have $\rho(\mathbf{x}, \mathbf{0}) \leq \rho(\mathbf{x}, \mathbf{x}_0) + \rho(\mathbf{x}_0, \mathbf{0}) = N + b =: P$, by the Triangle Inequality. Therefore, A is a subset of the cube $[-P, P]^n$, which is compact by Theorem on slide 19, since each $[-P, P] \subset \mathbb{R}$ is compact (by Corollary slide 35).

Since $A \subset [-P, P]^n$ is closed, A is compact by Theorem on slide 9. ■

Warning

The Heine-Borel Theorem does not state that "compact" and "closed and bounded" are equivalent in any metric space. For example, in the Banach space $(\ell^2, \|\cdot\|_2)$ of all square summable sequences of real numbers, the set $C = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \ldots\} = \{(1,0,0,\ldots),(0,1,0,\ldots),(0,0,1,0,\ldots),\ldots\}$ is closed and bounded, but not compact. Consider, for example, the open covering $\left\{B_d\left(\mathbf{e}_i, \sqrt{2}/2\right) \mid i \in \mathbb{N}\right\}$ of disjoint balls centered at the elements of set C has no proper subcover and so no finite subcover.

The space ℓ^2 is infinite dimensional and in fact, the Heine-Borel Theorem only holds in certain finite dimensional spaces. In particular, the closed unit ball is compact in a normed linear space if and only if the space is finite dimensional (see last section).

Extreme Value Theorem

Theorem (Extreme Value Theorem)

Let $f : X \to Y$ be continuous, where Y is an ordered set in the order topology. If X is compact, then there exists points $c, d \in X$ such that $f(c) \leq f(x) \leq f(d)$ for all $x \in X$.

Proof

Since f is continuous and X is compact, then by Theorem of slide 14, $A = f(X)$ is compact.
Assume that A has no largest element. Then the collection $\{(-\infty, a) \mid a \in A\}$ is an open covering of A and so has some finite subcover $\{(-\infty, a_1), (-\infty, a_2), \ldots, (-\infty, a_n)\}$. Let $a_i = \max\{a_1, a_2, \ldots, a_n\}$. Then $a_i \in A$ but a_i is not covered by the subcollection, a contradiction. So A does in fact have a largest element M. Similarly, A has a least element m. Then there are $c, d \in X$ such that $f(c) = m \leq f(x) \leq M = f(d)$ for all $x \in X$. ■

The Uniform Continuity Theorem

Before we prove this result we need two definitions and two preliminary lemmas.

Definition

Let (X, d) be a metric space. Let A be a nonempty subset of X. For each $x \in X$ define the distance from point x to set A as

$$d(x, A) = \inf\{d(x, a) \mid a \in A\}.$$

Lemma

Let (X, d) be a metric space and A a fixed subset of X. Then $D : X \to \mathbb{R}$ defined as $D(x) = d(x, A)$ is continuous.

Proof

Let $x \in X$ and $\varepsilon > 0$. Let $\delta = \varepsilon$. If $y \in X$ and $d(x, y) < \delta$ then for $a \in A$ we have $d(x, A) \le d(x, a) \le d(x, y) + d(y, a)$ (by the previous definition and the Triangle Inequality). Hence

$$d(x, A) - d(x, y) \le d(y, a), \quad \text{for all } a \in A$$

which implies that $d(x, A) - d(x, y) \le \inf_{a \in A} \{d(y, a)\} = d(y, A)$ or $d(x, A) - d(y, A) \le d(x, y) < \varepsilon$. Similarly (interchanging x and y),

$$d(y, A) - d(x, A) \le d(x, y) < \varepsilon$$

so $|d(x, A) - d(y, A)| = |D(x) - D(y)| < \varepsilon$. Hence D is continuous. ■

Lemma (The Lebesgue Number Lemma)

Let $\mathcal{A}$ be an open covering of metric space (X, d). If X is compact, there is a $\delta = \delta(\mathcal{A}) > 0$ such that for each subset B of X having diameter less than δ, there exists an element of $\mathcal{A}$ containing B. The number $\delta > 0$ is called a Lebesgue number for the covering $\mathcal{A}$.

"each small enough subset is included in one open set of the covering"

Proof

Let $\mathcal{A}$ be an open covering of X. If X itself is an element of $\mathcal{A}$ then any $\delta > 0$ is a Lebesgue number for $\mathcal{A}$, so without loss of generality, $X \notin \mathcal{A}$. Since X is compact, there is a finite subcollection $\{A_1, A_2, \ldots, A_n\} \subset \mathcal{A}$ that covers X. Set $C_i = X \backslash A_i$ for $i = 1, 2, \ldots, n$ and define $f : X \to \mathbb{R}$ as

$$f(x) = \frac{1}{n} \sum_{i=1}^{n} d(x, C_i)$$

(the average of the $d(x, C_i)$).

Proof (continued)

For arbitrary $x \in X$, we have $x \in A_i$ for some i. Since A_i is open and (X, d) is a metric space, for some $\varepsilon > 0$, $B_d(x, \varepsilon) \subset A_i$ and so $d\,(x, C_i) \geq \varepsilon$ and so $f(x) \geq \varepsilon/n > 0$. That is, f is strictly positive on X.

Since f is continuous by the previous lemma (and the fact that a sum of real valued continuous functions is continuous) and X is compact by hypothesis, by the Extreme Value Theorem f attains a minimum value $\delta > 0$ on X (this is where the fact that f is positive is used).

....

Proof (end)

Let B be a subset of X of diameter less than δ and let $x_0 \in B$. Then $B \subset B_d(x_0, \delta)$.

Now

$$\begin{aligned}\delta &\leq f(x_0) \quad \text{since } \delta \text{ is the minimum of } f \text{ on } X\\ &= \frac{1}{n}\sum_{i=1}^{n} d(x_0, C_i)\\ &\leq d(x_0, C_M)\end{aligned}$$

where $d(x_0, C_M)$ is the largest of the numbers $d(x_0, C_i)$. Then the δ-neighborhood $B_d(x_0, \delta)$ of x_0 is contained in the element $A_M = X \backslash C_M \in \mathcal{A}$. Since $B \subset X$ of diameter less than δ is arbitrary, then δ is a Lebesgue number for $\mathcal{A}$. ∎

Definition

A function f for the metric space (X, d_X) to the metric space (Y, d_Y) is uniformly continuous if given $\varepsilon > 0$ there is $\delta > 0$ such that for every pair $x_0, x_1 \in X$ we have

$$d_X(x_0, x_1) < \delta \Longrightarrow d_Y(f(x_0), f(x_1)) < \varepsilon.$$

Theorem (Uniform Continuity Theorem)

Let $f : X \to Y$ be a continuous map of the compact metric space (X, d_X) to the metric space (Y, d_Y). Then f is uniformly continuous on X.

Proof

Let $\varepsilon > 0$. Consider the open covering of Y of $\left\{B_{d_Y}(y, \varepsilon/2) \mid y \in Y\right\}$. Since f is continuous, each $f^{-1}\left(B_{d_Y}(y, \varepsilon/2)\right)$ is open. Let $\mathcal{A}$ be the open covering of X given by $\mathcal{A} = \left\{f^{-1}\left(B_{d_Y}(y, \varepsilon/2)\right) \mid y \in Y\right\}$. (Note that $\mathcal{A} = \mathcal{A}(\varepsilon)$)

Since X is a compact metric space, then $\mathcal{A}$ has a Lebesgue number δ by The Lebesgue Number Lemma. Then if $x_1, x_2 \in X$ with $d_X(x_1, x_2) < \delta$, the two-point set $\{x_1, x_2\}$ has diameter less than δ so that $\{x_1, x_2\}$ is a subset of some element of $\mathcal{A}$ and so $\{f(x_1), f(x_2)\}$ lies in some $B_{d_Y}(y, \varepsilon/2)$. Then $d_Y(f(x_1), f(x_2)) < \varepsilon$. So f is uniformly continuous. ■

Isolated points and uncountability

We now present a topological property concerning the cardinality of a space. This result will allow us to show that $\mathbb{R}$ is uncountable.

Definition

If X is a space, a point $x \in X$ is an isolated point of X if the one-point set $\{x\}$ is open in X.

Theorem

Let X be a nonempty compact Hausdorff space. If X has no isolated points, then X is uncountable.

Proof

We follow the two-step proof of Munkres.

- **Step 1**. Let U be a nonempty open set of X and let $x \in X$. Let $y \in U$ where $y \neq x$ (this is possible since X has no isolated points and so if $x \in U$ then $|U| \neq 1$; if $x \notin U$ this is possible since $U \neq \varnothing$). Since X is Hausdorff, there are disjoint open W_1 and W_2 with $x \in W_1$ and $y \in W_2$. Let $V = W_2 \cap U$. Then V is open, $y \in V, v \neq \varnothing$, and $V \subset U$. Since W_1 is a neighborhood of x which does not intersect V, then x is not a limit point of V (by the definition of "limit point"). By Theorem of slide 58, Chap. 2, $\bar{A} = A \cup A'$ (where A' is the set of limit points of A) so $x \notin \bar{V}$. Therefore, for any open set U and any $x \in X$ there is a nonempty set $V \subset U$ such that $x \notin \bar{V}$.

Proof (continued)

- **Step 2**. Suppose $f : \mathbb{N} \to X$. Denote $x_n = f(x_n)$. By Step 1 applied to nonempty open set $U = X$, there is a nonempty open set $V_1 \subset X$ such that $x_1 \notin \bar{V}_1$. Inductively, define V_{n+1} given V_n by applying Step 1 to nonempty open set $V_n = U$ to produce nonempty open set $V_{n+1} \subset U = V_n$ such that $x_{n+1} \notin \bar{V}_{n+1}$. Then we have the sequence of nonempty, nested, closed sets $\bar{V}_1 \supset \bar{V}_2 \supset \bar{V}_3 \supset \cdots$. Since X is compact, by Corollary slide 24 there is some $x \in \cap_{m=1}^{\infty} \bar{V}_m$. Notice that for each x_n, we have $x_n \notin \bar{V}_n$ and so $x_n \notin \cap_{m=1}^{\infty} \bar{V}_m$ for all $n \in \mathbb{N}$. So x is different from all x_n and so there is no $n \in \mathbb{N}$ such that $f(n) = x$. Therefore, arbitrary function $f : \mathbb{N} \to X$ is not surjective (onto). Hence, $|\mathbb{N}| < |X|$ and X is uncountable. $\blacksquare$

Since a closed interval $[a, b] \subset \mathbb{R}$ is a compact set in $\mathbb{R}$ by Corollary on slide 35 and has no isolated points, then by the preceding Theorem, $[a, b]$ is not countable. We therefore have the following.

Corollary

Every closed and bounded interval $[a, b] \subset \mathbb{R}$ is uncountable.

Outline

Limit Point Compactness

We introduce two properties equivalent to compactness for metrizable spaces. One of the properties is stronger than compactness in a more general setting.

Definition

A space X is limit point compact if every infinite subset of X has a limit point.

The term "limit point compact" is due to Munkres. Munkres comments that the property is sometimes also called "Fréchet compactness" or the "Bolzano-Weierstrass property". Recall that the Bolzano-Weierstrass Theorem states that a bounded infinite set of real numbers (or elements of $\mathbb{R}^n$) must have a limit point.

Theorem

Compactness implies limit point compactness, but not conversely.

Proof

Let X be compact and let $A \subset X$. We prove the (logically equivalent) contrapositive of the claim: If A has no limit point, then A must be finite. We start with two observations:

- Since $A \subset X$ has no limit points then for each $a \in A$, since a is not a limit point of A, there is a neighborhood U_a of a such that U_a intersects A in the point a only.
- A is necessarily closed because since it has no limit points it trivially contains them all.

Proof (continued)

Now, the space X is covered by the open set $X \backslash A$ and each open U_a. Since X is compact, it can be covered by finitely many of these sets. Since $X \backslash A$ does not intersect A, each open set U_a contains only one point of A and so the set A is necessarily finite.

So if X is compact then it is limit point compact. The fact that the converse does not necessarily hold is given in the following example. ∎

Example

Let $Y = \{y_1, y_2\}$ and let the topology on Y consist of $\varnothing$ and Y. Consider $X = \mathbb{N} \times Y$ with the product topology where $\mathbb{N}$ has the discrete topology. Then for $A \subset X, A \neq \varnothing, A$ has an element of the form (n, y_i). Any open set containing (n, y_i) also contains the point $\left(n, y_j\right)$ where $j = 3 - i$, and so every nonempty $A \subset X$ has a limit point and so X is limit point compact. However, X is not compact since the covering of X by the open sets $U_n = \{n\} \times Y$ for $n \in \mathbb{N}$ has no subcollection covering X (and so no finite subcollection covering X).

Sequential compactness

Definition

Let X be a topological space. If $\{x_n\}_{n=1}^{\infty}$ is a sequence of points in X and if $n_1 < n_2 < \cdots < n_i < \cdots$ is an increasing sequence of natural numbers, then the sequence $\{y_i\}_{i=1}^{\infty}$ defined as $y_i = x_{n_i}$ is a subsequence of the sequence $\{x_n\}$.
The space X is sequentially compact if every sequence of points of X has a convergent subsequence.

We now show that each of three three types of compactness are the same in metrizable spaces. We follow Munkres' proof, but break it into pieces, including two preliminary lemmas.

Lemma

Let X be metrizable. If X is also sequentially compact then the conclusion of the Lebesgue Number Lemma holds for X.

Proof

Let $\mathcal{A}$ be an open covering of X. Assume there is no Lebesgue number for open covering $\mathcal{A}$. That is, assume there is no $\delta > 0$ such that each set of diameter less than δ has an element of $\mathcal{A}$ containing it. So for each $n \in \mathbb{N}$, there is a set of diameter less than $1/n$ that is not contained in any element of $\mathcal{A}$. Let C_n be such a set. Choose $x_n \in C_n$ for each $n \in \mathbb{N}$. Since X is hypothesized to be sequentially compact, then some subsequence $\{x_{n_i}\}$ of $\{x_n\}$ which converges, say to point a. Now $a \in A$ for some $A \in \mathcal{A}$. Since A is open and X is metrizable, there is $\varepsilon > 0$ such that $B(a, \varepsilon) \subset A$.

If i is large enough so that $1/n_i < \varepsilon/2$, then the set C_{n_i} has diameter less than $1/n_i < \varepsilon/2$ and so $C_{n_i} \subset B\left(x_{n_i}, \varepsilon/2\right)$.

Proof (continued)

If i is also chosen large enough so that $d\left(x_{n_i}, a\right) < \varepsilon/2$ (which can be done since $\left\{x_{n_i}\right\} \to a$), then $C_{n_i} \subset B(a, \varepsilon) \subset A$.
But this contradicts the assumption that $\mathcal{A}$ has no Lebesgue number (and the implication of that assumption that such C_n exists which is not a subset of some element of $\mathcal{A}$). Therefore, there is a Lebesgue number for $\mathcal{A}$. Since $\mathcal{A}$ is an arbitrary open covering of X, then X satisfies the conclusion of the Lebesgue Number Lemma. ■

Lemma

Let X be metrizable. If X is also sequentially compact, then for all $\varepsilon > 0$ there exists a finite covering of X by open ε-balls.

Proof

Assume that, to the contrary of the claim, there is $\varepsilon > 0$ such that X cannot be covered by finitely many ε-balls. Construct sequence $\{x_n\}$ as follows: First, let $x_1 \in X$ be any point in X. By assumption, $B(x_1, \varepsilon)$ is not all of X, so there is $x_2 \in X \backslash B(a_1, \varepsilon)$. Inductively, let $x_{n+1} \in X \backslash (B(x_1, \varepsilon) \cup B(x_2, \varepsilon) \cup \cdots \cup B(x_n, \varepsilon))$; such x_{n+1} exists since the n ε-balls are assumed to not cover X. By construction, $d(x_{n+1}, x_i) \geq \varepsilon$ for all $i = 1, 2, \ldots, n$. So any $\varepsilon/2$-ball in X can contain either one or no elements of the sequence $\{x_n\}$. Hence $\{x_n\}$ can have no convergent subsequence. But this contradicts the sequential compactness of X. So the claim holds.

Theorem

Let X be a metrizable space. Then the following are equivalent:

(1) *X is compact.*

(2) *X is limit point compact.*

(3) *X is sequentially compact.*

Proof

(1) $\Rightarrow$ (2) : This follows from Theorem on slide 54.

(2) $\Rightarrow$ (3) : Suppose X is limit point compact. Let $\{x_n\}$ be a sequence of points of X. Consider the set $A = \{x_n \mid n \in \mathbb{N}\}$. If set A is finite, then there is at least one point x such that $x = x_n$ for infinitely many values $n \in \mathbb{N}$. In this case, $\{x_n\}$ has a subsequence that is constant and hence convergent. On the other hand, if A is infinite, then A has a limit point x since X is hypothesized to be limit point compact.

Proof (continued)

We define a subsequence of $\{x_n\}$ converging to the limit point x as follows: Let n_1 be such that $x_{n_1} \in B(x, 1)$. Inductively define n_{i+1} in terms of n_i by letting n_{i+1} be such that $B(x, 1/i)$ contains $x_{n_{i+1}}$ and $n_{i+1} > n_i$ (such n_{i+1} exists since $B(x, 1/i)$ contains infinitely many points of A). Then the subsequence $\left\{x_{n_i}\right\}_{i=1}^{\infty}$ converges to x. Since $\{x_n\}$ is an arbitrary sequence in X, then X is sequentially compact.

$(3) \Rightarrow (1)$: Let $\mathcal{A}$ be an open covering of sequentially compact metrizable X. Then by the lemma in slide 58, the covering $\mathcal{A}$ has a Lebesgue number δ. Let $\varepsilon = \delta/3$. By the lemma on page 60, there is a finite covering of X with open ε-balls. Each of these balls has diameter at most $2\delta/3 < \delta$ and so each ball lies in an element of $\mathcal{A}$. Choose one element of $\mathcal{A}$ for each of these finite number of ε-balls and, since the ε-balls cover X, then finite subcollection of $\mathcal{A}$ covers X. Since open covering $\mathcal{A}$ is arbitrary, then X is compact. ∎

Outline

Local Compactness

Definition

A topological space X is locally compact at point x if there is some compact subspace X of X that contains a neighborhood of x. If X is locally compact at each of its points, set X is locally compact.

Example

$\mathbb{R}$ is locally compact since $x \in \mathbb{R}$ lies in neighborhood $(x-1, x+1)$ which is in the compact space $[x-1, x+1]$. In the tutorials you will show that $\mathbb{Q}$ is not locally compact.

Example

Similar to the argument of $\mathbb{R}$, we have that $\mathbb{R}^n$ is locally compact. For $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{x}$ is in the basis element

$$(x_1 - 1, x_1 + 1) \times (x_2 - 1, x_2 + 1) \times \cdots \times (x_n - 1, x_n + 1)$$

which in turn is contained in compact subspace

$$[x_1 - 1, x_1 + 1] \times [x_2 - 1, x_2 + 1] \times \cdots \times [x_n - 1, x_n + 1] .$$

Example

Every simple ordered set X having the least upper bound property is locally compact since a basis element for X is of the form (a, b), $[a, b)$, or $(a, b]$. The closure of any basis element is then a closed interval which, by Corollary slide 35, is compact. So X is locally compact (see the previous example for details relating basis elements to neighborhoods).

Example

$\mathbb{R}^\omega = \mathbb{R} \times \mathbb{R} \times \cdots$ under the product topology is not locally compact. Recall that by basis elements for the product topology are of the form

$$B = (a_1, b_1) \times (a_1, b_2) \times \cdots \times (a_n, b_n) \times \mathbb{R} \times \mathbb{R} \times \cdots .$$

If C is a compact subspace of $\mathbb{R}^\omega$ that contains $\mathbf{x} \in \mathbb{R}^\omega$ and there is a neighborhood of $\mathbf{x}$ in C, then the neighborhood contains a basis element of the form of B. But then

$$\bar{B} = [a_1, b_1] \times [a_1, b_2] \times \cdots \times [a_n, b_n] \times \mathbb{R} \times \mathbb{R} \times \cdots$$

is a closed subspace of C and so would be compact by Theorem on slide 9. But $\bar{B}$ is not compact (consider an open cover which requires all of the open sets to cover one of the $\mathbb{R}$ components of B). So C cannot be compact and $\mathbb{R}^\omega$ is not locally compact.

Local compactness in metric and Banach spaces

The last example shows that metrizability does not suffice to ensure local compactness (see Theorem 20.5 in [Munkres] for a proof that $\mathbb{R}^\omega$ with the product topology is metrizable).

The same statement applies to normed spaces. Indeed, the goal of the next paragraphs is showing that local compactness in the context of normed spaces is equivalent to being finite-dimensional (in the sense of linear algebra).

We start by showing several important properties of normed spaces.

Proposition

Let $(E, \|\cdot\|)$ *be a normed space,* $x \in E$*, and* $\epsilon > 0$*. Then,*

$$\overline{B_{\|\cdot\|}(x, \epsilon)} = B_{\|\cdot\|}[x, \epsilon].$$

Proof

We proceed by double inclusion. Take first $y \in B_{\|\cdot\|}[x, \epsilon]$, that is, $\|x - y\| \leq \epsilon$. The triangle inequality implies that the segment $\{xt + (1-t)y \mid t \in [0, 1]\}$ is included in $B_{\|\cdot\|}[x, \epsilon]$ (closed ball is convex). Consider now $\epsilon' > 0$ small enough so that $t_0 = \epsilon'/2 \|x - y\| < 1$. Let $z = xt_0 + (1 - t_0)y$ and consider $B_{\|\cdot\|}(y, \epsilon')$. Then, it is easy to check that $\|z - y\| = \epsilon'/2 < \epsilon'$ and hence $z \in B_{\|\cdot\|}(y, \epsilon')$. Additionally, it can be verified that $\|x - z\| + \|z - y\| = \|x - y\|$ and hence

$$\|x - z\| = \|x - y\| - \|z - y\| = (1 - t_0)\epsilon < \epsilon,$$

which proves that $z \in B_{\|\cdot\|}(x, \epsilon)$ and that $B_{\|\cdot\|}(x, \epsilon) \cap B_{\|\cdot\|}(y, \epsilon') \neq \emptyset$.

Proof (continued)

Since given $\epsilon > 0$ this identity holds for any $\epsilon' > 0$ this implies that $y \in \overline{B_{\|\cdot\|}(x, \epsilon)}$ and hence that

$$B_{\|\cdot\|}[x, \epsilon] \subseteq \overline{B_{\|\cdot\|}(x, \epsilon)}.$$

Conversely, let $y \in \overline{B_{\|\cdot\|}(x, \epsilon)}$. Then for any $\epsilon' > 0$ one has that $B_{\|\cdot\|}(x, \epsilon) \cap B_{\|\cdot\|}(y, \epsilon') \neq \emptyset$. Let z be an element in that intersection. Then,

$$\|y - x\| \leq \|y - z\| + \|z - x\| < \epsilon' + \epsilon.$$

Since this inequality holds true for all $\epsilon' > 0$, we conclude that $\|y - x\| \leq \epsilon$ and hence that $y \in B_{\|\cdot\|}[x, \epsilon]$. ■

Proposition

Let $(E, \|\cdot\|)$ *be a normed space,* $y \in E$*, and* $\lambda \in \mathbb{R}$ *such that* $\lambda \neq 0$*. The translation and dilation maps* $T_y : E \longrightarrow E$ *and* $D_\lambda : E \longrightarrow E$ *defined by*

$$T_y(x) = x + y, \qquad D_\lambda(x) = \lambda x, \quad \text{for all } x \in E,$$

are homeomorphisms such that

$$T_y\left(B_{\|\cdot\|}(x, \epsilon)\right) = B_{\|\cdot\|}(x + y, \epsilon) \quad \text{and} \quad D_\lambda\left(B_{\|\cdot\|}(x, \epsilon)\right) = B_{\|\cdot\|}(\lambda x, \lambda\epsilon)$$

Proposition

Let $(E, \|\cdot\|)$ be a finite-dimensional normed space of dimension $n \in \mathbb{N}$. Then $(E, \|\cdot\|)$ is homeomorphic to $(\mathbb{R}^n, \|\cdot\|_1)$.

Important observation

This proposition guarantees that all the norms on a finite dimensional vector space induce the same topology.

Proof

Let $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ be a basis of E and let $T : E \longrightarrow \mathbb{R}^n$ be the linear map defined by

$$\begin{array}{rccc} T: & E & \longrightarrow & \mathbb{R}^n \\ & v = \sum_{i=1}^n x_i \mathbf{e}_i & \longmapsto & (x_1, \ldots, x_n). \end{array}$$

Since E and $\mathbb{R}^n$ have the same dimension then T is necessarily a linear isomorphism. It only remains to be proved that T and its inverse T^{-1} are continuous using the criteria in Theorem on slide 21 of Chapter 4.

Proof (continued)

We start by showing that T^{-1} is bounded. For any $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$ we have

$$\left\|T^{-1}(x)\right\| = \left\|\sum_{i=1}^{n} x_i \mathbf{e}_i\right\| \leq \sum_{i=1}^{n} |x_i| \, \|\mathbf{e}_i\|$$

$$\leq \max_{i\in\{1,\ldots,n\}} \{\|\mathbf{e}_i\|\} \sum_{i=1}^{n} |x_i| = \max_{i\in\{1,\ldots,n\}} \{\|\mathbf{e}_i\|\} \, \|x\|_1 ,$$

which implies that $\left|\left|\left|T^{-1}\right|\right|\right| \leq \max_{i\in\{1,\ldots,n\}} \{\|\mathbf{e}_i\|\} < \infty$ and T^{-1} is hence continuous.

We prove that T is continuous by contradiction. Suppose T is not continuous at zero. Let $(y_n)_{n\in\mathbb{N}}$ be a sequence in E such that $y_n \to 0$ and $(T(y_n))_{n\in\mathbb{N}}$ does not converge to 0. This implies that there exists $\epsilon > 0$ and $N(\epsilon) \in \mathbb{N}$ such that for all $n > N(\epsilon)$ we have that $\|T(y_n)\|_1 > \epsilon$.

Proof (continued)

This allows us to define a new sequence $z_n = y_n / \|T(y_n)\|_1$. It is obvious that $z_n \longrightarrow 0$ and that $\|T(z_n)\|_1 = 1$. Since the set of vectors in $\mathbb{R}^n$ with norm one is compact (by Heine-Borel) then it is limit point compact and hence $(T(z_n))_{n\in\mathbb{N}}$ has a convergent subsequence $(T(z_{\alpha(n)}))_{n\in\mathbb{N}}$ with limit y such that $\|y\| = 1$.

Now, since T^{-1} is continuous $z_{\alpha(n)} \longrightarrow T^{-1}(y)$. Since $z_n \longrightarrow 0$ then $T^{-1}(y) = 0$ which implies $y = 0$ since T^{-1} is injective. This is not possible since $\|y\| = 1$. This is a contradiction. ■

Local compactness and the closed unit ball

Proposition

Let $(E, \|\cdot\|)$ be a normed space. Then $(E, \|\cdot\|)$ is locally compact if and only if the closed unit ball $B_{\|\cdot\|}[0,1]$ is compact.

Proof

Suppose that $(E, \|\cdot\|)$ is locally compact. Then, for each $x \in E$ there is a compact subspace C that contains an open neighborhood U_x of x. This implies that there exists $\epsilon > 0$ such that $B_{\|\cdot\|}(x, \epsilon) \subset U_x \subset C$. Using the result in slide 68 we have that $B_{\|\cdot\|}[x, \epsilon] = \overline{B_{\|\cdot\|}(x, \epsilon)} \subset \overline{C} = C$. We have that $\overline{C} = C$ because C is Hausdorff. Hence, since by these relations $B_{\|\cdot\|}[x, \epsilon]$ is a closed subset of the compact space C, then it is compact. We can now translate and dilate this closed ball using the homeomorphisms in slide 70 to show that then $B_{\|\cdot\|}[0,1]$ is compact.

Proof (continued)

Conversely, if $B_{\|\cdot\|}[0,1]$ is compact, then using the homeomorphisms in slide 70 we can conclude that for any $x \in E$ the compact set $B_{\|\cdot\|}[x,1]$ contains the open neighborhood $B_{\|\cdot\|}(x,1)$ of x. ■

Finite dimensionality and local compactness

Theorem

A normed space $(E, \|\cdot\|)$ is locally compact if and only if it is finite dimensional.

Proof

($\Longleftarrow$) If $(E, \|\cdot\|)$ is finite dimensional, by the Proposition in slide 71 it is homeomorphic to $(\mathbb{R}^n, \|\cdot\|_1)$. At the same time $(\mathbb{R}^n, \|\cdot\|_1)$ is locally compact because of the previous proposition and the Heine-Borel Theorem. Since local compactness is a topological property then $(E, \|\cdot\|)$ is necessarily locally compact.

($\Longrightarrow$) Suppose that $(E, \|\cdot\|)$ is locally compact. By the previous proposition, $B_{\|\cdot\|}[0,1]$ is compact. This implies that it can be covered by a finite number of open balls of radius $1/2$, that is,

$$B_{\|\cdot\|}[0,1] \subset \cup_{i=1}^{n} B_{\|\cdot\|}(x_i, 1/2).$$

Proof (continued)

We proceed in three steps:

(1) **Step 1: the linear space F.** Consider now the linear subspace F of E spanned by the vectors $\{x_1, \ldots, x_n\}$ which has dimension $k \leq n$. The restriction of the norm in E to F makes it into a k-dimensional normed space that, by the proposition in slide 71, is homeomorphic to $(\mathbb{R}^k, \|\cdot\|_1)$. Since $(\mathbb{R}^k, \|\cdot\|_1)$ is complete then so is F and then it is necessarily closed (complete subsets of metric spaces are always closed). We will obtain the proof by showing that $F = E$ necessarily.

(2) **Step 2: the distance d.** By contradiction, assume that there exists $v \in E$ such that $v \notin F$. Given that $F = \overline{F}$ we have that the number

$$d = \inf \{\|v - e\| \mid e \in F\}$$

is strictly positive.

Proof (continued)

(3) **Step 3: the compact space $B_{\|\cdot\|}[v,r] \cap F$.** Take now $r > 0$ such that $B_{\|\cdot\|}[v,r] \cap F \neq \emptyset$. Given that $B_{\|\cdot\|}[v,r] \cap F$ is a closed and bounded subset of the finite dimensional vector space F, by the Heine-Borel Theorem it is necessarily compact. Notice now that

- $d = \inf \{\|v - e\| \mid e \in F\} = \inf \left\{\|v - e\| \mid e \in B_{\|\cdot\|}[v,r] \cap F\right\}$.
- By Lemma slide 42, the map $e \in B_{\|\cdot\|}[v,r] \cap F \longmapsto \|v - e\|$ is continuous.

These two points imply that there exists $e_0 \in B_{\|\cdot\|}[v,r] \cap F$ such that

$$d = \|v - e_0\|.$$

Proof (continued)

(4) **Step 4: the contradiction**: we can deduce the following two statements:

- $\dfrac{v-e_0}{\|v-e_0\|}$ has norm one and hence belongs to $B_{\|\cdot\|}[0,1]$ which implies that for one of the points x_i (of Step 1), $\left\|\dfrac{v-e_0}{\|v-e_0\|}-x_i\right\|<\dfrac{1}{2}$, (recall $B_{\|\cdot\|}[0,1]\subset\cup_{i=1}^{n}B_{\|\cdot\|}(x_i,1/2)$) and

$$\|v-e_0-\|v-e_0\|\,x_i\|<\frac{1}{2}\|v-e_0\|=\frac{d}{2}.$$

- Since $e_0, x_i\in F$ and F is a vector space, then $e_0+\|v-e_0\|\,x_i\in F$ and hence its distance to F is greater or equal to d, that is

$$\|v-e_0-\|v-e_0\|\,x_i\|\geq d.$$

which contradicts the previous point. ■

Outline

References I

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